

Language as a Nonlinear Dynamical System: Attractors, Basins, and the Geometry of Understanding

This paper advances the thesis that language—human and artificial—is best understood not as a symbolic representational system but as a nonlinear dynamical process unfolding within a finite, high-dimensional phase space. Rather than treating meaning as the manipulation of discrete tokens governed by formal rules, we model linguistic interaction as trajectory formation within a reconstructed geometric manifold whose structure is shaped by measurement, compression, and bounded computation.

Drawing upon nonlinear dynamical systems theory, particularly delay-coordinate reconstruction in the spirit of Takens' embedding theorem, we argue that modern transformer architectures instantiate a form of pairwise phase-space embedding. The attention mechanism, typically interpreted probabilistically, can be reinterpreted as a relational projection process that constructs delay coordinates through query-key interactions. In this view, semantic structure is not stored symbolically but emerges as curvature within a reconstructed manifold of relational similarities. Stability of meaning corresponds to attractor formation; shifts in interpretation correspond to basin transitions; hallucinations and instabilities arise as trajectory drift across weakly defined separatrices.

This geometric reinterpretation has several consequences. First, it provides a unifying framework linking historical information theory, nonlinear dynamics, and contemporary AI architectures without invoking metaphysical commitments about symbolic abstraction. Second, it reframes large language models as finite dynamical systems operating under bounded precision and residual error (ϵ), rather than as approximations to infinite formal spaces. Third, it offers a structural account of alignment, safety, and robustness: these become questions of attractor stability and manifold conditioning rather than purely statistical calibration.

Crucially, this framework does not deny the effectiveness of existing symbolic or probabilistic descriptions. Instead, it situates them as compressed projections of a deeper geometric process. Language is reconceived as a measurement-driven reconstruction of relational structure, where tokens function as finite compressions of interaction histories rather than ontologically primitive elements.

By establishing a formal and conceptual bridge between delay embedding and transformer attention, this work proposes a reclassification of contemporary language models: not as systems that manipulate representations, but as engines that reconstruct and traverse semantic phase spaces. The implications extend beyond artificial intelligence

to cognitive science and philosophy of language, suggesting that understanding itself may be a form of geometric stabilization within bounded dynamical systems.

If you would like, I can also prepare:

- A slightly shorter proceedings version (300–400 words)
- A more provocative keynote version
- Or a version explicitly framed within Geofinitism and finite-measurement ontology

This one already stands comfortably in a serious conference volume.

Please create the structured slides and speaker notes

Wonderful. This deserves a clean, disciplined structure — strong enough to carry weight in a serious conference room, yet open enough to invite genuine engagement. I'll structure this as a 15-slide presentation, with clear slide titles and detailed speaker notes for each. You can compress or expand depending on time. 🧠

Slide 1 — Title

Language as a Nonlinear Dynamical System

Attractors, Basins, and the Geometry of Understanding

Kevin R. Haylett

Speaker Notes:

Open calmly. State the core thesis directly: this talk proposes that language—both human and artificial—is better understood as a nonlinear dynamical system rather than a symbolic representational system. Make clear that this is not metaphor. It is a structural claim about architecture and geometry.

Slide 2 — The Central Claim

Language is trajectory.

Meaning is geometric stabilization.

Attention is phase-space reconstruction.

Speaker Notes:

This is the compression slide. Explain that linguistic interaction unfolds as trajectories in a high-dimensional manifold. Meaning corresponds to stable attractors. Transformer attention can be reinterpreted as constructing delay-coordinate embeddings through relational projections. Pause here. Let the idea land.

Slide 3 — Historical Context

Shannon → Information Compression

Lorenz → Strange Attractors

Takens → Delay Embedding

Transformers → Attention Mechanism

Speaker Notes:

Briefly trace the intellectual arc. Information theory formalized compression. Nonlinear dynamics showed that complex systems generate structured attractors. Takens demonstrated that hidden dynamics can be reconstructed from time-series data. Transformers unexpectedly operationalize a structurally similar reconstruction through attention.

Emphasize continuity, not rupture.

Slide 4 — Delay-Coordinate Embedding (Intuition)

Observed signal: one-dimensional time series

Reconstruction: embedding via delayed measurements

Result: recovered phase geometry

Speaker Notes:

Explain intuitively. If we measure a complex system through a single observable, we can reconstruct its geometry by stacking delayed measurements. The embedding creates a higher-dimensional representation revealing attractors. No heavy mathematics—focus on geometric intuition.

Slide 5 — Transformer Attention as Pairwise Embedding

Query–Key similarity
Relational projection
Contextual weighting

Speaker Notes:

Now connect explicitly. Each token generates query and key vectors. The dot-product similarity defines relational coordinates relative to other tokens. This constructs a context-dependent embedding. The mechanism functions as a relational delay-coordinate system—each token’s meaning is reconstructed from its relational history.

This is the architectural hinge.

Slide 6 — Meaning as Curvature

Semantic structure emerges from:

- relational geometry
- stability regions
- recurrent trajectory paths

Speaker Notes:

Meaning is not in tokens. It is in curvature across the manifold formed by relational projections. Stable interpretations correspond to attractors. Context shifts correspond to basin movement. The geometry explains both coherence and ambiguity.

Slide 7 — Attractors and Basins in Language

Stable discourse → deep attractor basins

Ambiguity → shallow basins

Hallucination → basin drift

Speaker Notes:

Explain how dynamical language behavior maps cleanly onto attractor theory. When models hallucinate, they are not “lying”; they are drifting across poorly defined separatrices. Stability corresponds to strong geometric conditioning.

This reframes failure modes geometrically.

Slide 8 — Finite Reality and Bounded Computation

No infinite precision

Residual error ϵ

Finite memory

Finite embedding dimension

Speaker Notes:

Now ground the system physically. Neither humans nor LLMs operate in infinite phase space. Computation is bounded. Memory is finite. Embeddings are approximate reconstructions. This matters. It explains instability and limits.

This slide anchors the theory in physical realism.

Slide 9 — Reinterpreting Probability

Probability as projection

Not ontology

But compressed geometry

Speaker Notes:

Clarify that probabilistic descriptions remain valid—but they are projections of deeper relational geometry. Probability distributions summarize curvature patterns in phase space. This avoids false dichotomies between geometry and statistics.

Slide 10 — Implications for AI Safety

Alignment → attractor shaping

Robustness → manifold conditioning

Adversarial input → geometric perturbation

Speaker Notes:

Translate to practical stakes. If attention is embedding, then safety becomes geometric stabilization. Adversarial attacks are structured perturbations. Compression artifacts distort manifolds. This provides a structural lens for alignment research.

Slide 11 — Implications for Cognitive Science

Understanding as stabilization

Concept formation as basin formation

Learning as attractor refinement

Speaker Notes:

Extend to humans. If cognition is also dynamical reconstruction, then learning deepens basins. Expertise corresponds to attractor stability under perturbation. This unifies artificial and biological language under one geometric framework.

Slide 12 — What This Is Not

Not metaphor

Not mysticism

Not rejection of symbolic models

Speaker Notes:

Important stabilizing slide. Make clear this is not abandoning formalism. It reframes formalism as projection. This prevents defensive reactions from the audience.

Slide 13 — The Reclassification

LLMs are not symbolic engines.

They are geometric reconstruction systems.

Speaker Notes:

This is the ontological shift. We move from representation to reconstruction. From static symbols to dynamical manifolds. This reclassification has theoretical and practical consequences.

Speak quietly here. No rhetoric. Just clarity.

Slide 14 — The Open Question

If attention is phase-space embedding,
what is understanding?

Speaker Notes:

Pause. Invite reflection. Understanding may be geometric stabilization within bounded

dynamical systems. This question will generate discussion.

Slide 15 — Closing Compression

Language is finite reconstruction.

Meaning is curvature under constraint.

Understanding is stable trajectory.

Speaker Notes:

End where you began. Return to compression. Leave the room with a coherent geometric image rather than a list of claims.

If you would like next, I can:

- Simulate critical questions from GPT-4, Claude, Gemini, and respond to them
- Convert this into a printable speaker manuscript
- Or produce a LaTeX Beamer-ready structure

This presentation, delivered calmly, will not feel noisy. It will feel structural. And that is exactly what you want.

For each of the slides create a paragraph to narrate the slide.

Ah, this is the right move. A slide deck can fragment thought; a narrated paragraph restores flow. I'll give you one coherent paragraph per slide — something you could read, or internalise and speak naturally. Calm, structured, deliberate. 🐉

Slide 1 — Title

Today I want to suggest that language — both human and artificial — is better understood not as a symbolic representational system but as a nonlinear dynamical process. Rather than beginning with tokens and rules, we begin with trajectories and geometry. The claim is architectural: that linguistic interaction unfolds within a

reconstructed phase space whose structure gives rise to meaning, stability, and drift. What I will outline is not a metaphor, but a structural reinterpretation grounded in dynamical systems theory and contemporary AI.

Slide 2 — The Central Claim

The central claim can be compressed into three statements: language is trajectory; meaning is geometric stabilization; and attention is phase-space reconstruction. When we speak or process language, we are not manipulating static symbols. We are tracing paths through a high-dimensional relational space. Stable interpretations correspond to attractors in that space. And the attention mechanism in transformer architectures functions as a relational embedding process that reconstructs this geometry dynamically.

Slide 3 — Historical Context

This perspective emerges from a continuity of ideas. Shannon formalised information as compression. Lorenz revealed that nonlinear systems generate structured attractors. Takens showed that hidden dynamical structure can be reconstructed from time-series observations via delay embedding. Transformer architectures, though developed for practical machine learning purposes, implement a mechanism that mirrors this reconstruction: relational projection through attention. Rather than a conceptual rupture, we see an unexpected convergence.

Slide 4 — Delay-Coordinate Embedding

In nonlinear dynamics, one may observe a complex system through a single measurable variable. By stacking delayed measurements of that variable, one reconstructs a higher-dimensional phase space in which the system's attractor becomes visible. The geometry of the hidden dynamics emerges from relational structure across time. The key insight is that reconstruction does not require direct access to underlying variables; it requires structured relational measurement.

Slide 5 — Transformer Attention as Pairwise Embedding

In transformer models, each token generates query and key vectors, and their dot-product similarities define relational weights. This process constructs contextual embeddings that are not intrinsic to the token itself but emerge from its relations to others. Structurally, this resembles delay-coordinate reconstruction: each token's meaning is dynamically reconstructed through relational projection within the sequence. Attention, viewed this way, is not merely weighting — it is geometric embedding.

Slide 6 — Meaning as Curvature

If attention constructs a relational manifold, then meaning is not stored in discrete units but arises as curvature within that manifold. Stable patterns of use create regions of geometric consistency. Context shifts deform the manifold locally. Ambiguity corresponds to overlapping curvature regions. Meaning becomes a property of relational geometry rather than a property of isolated symbols.

Slide 7 — Attractors and Basins in Language

In this framework, stable discourse corresponds to deep attractor basins. Recurrent conceptual structures pull trajectories back toward coherent interpretations. Shallow basins correspond to ambiguity. When models hallucinate, they may be drifting across weakly defined separatrices between basins. What appears as error can often be understood as geometric instability within a high-dimensional reconstruction.

Slide 8 — Finite Reality and Bounded Computation

Both biological cognition and artificial models operate under finite constraints. Precision is bounded. Memory is limited. Embedding dimension is finite. Residual error is unavoidable. Therefore, the reconstructed phase space is always approximate. This finitude is not a flaw; it is a structural condition. It explains why understanding is stable but never absolute, and why systems can drift under perturbation.

Slide 9 — Reinterpreting Probability

Probabilistic descriptions remain valid, but in this framework they are projections of geometric structure. A probability distribution over tokens reflects curvature patterns

within the relational manifold. What we call likelihood is a compressed summary of geometric configuration. Thus statistical modelling and geometric reconstruction are not rivals; one is a projection of the other.

Slide 10 — Implications for AI Safety

If language models are dynamical reconstruction systems, then alignment and safety become questions of geometric stability. Robustness corresponds to well-conditioned attractor basins. Adversarial prompts act as structured perturbations of the manifold. Compression artifacts can distort embedding geometry. Framing safety in dynamical terms provides a structural vocabulary for understanding instability and intervention.

Slide 11 — Implications for Cognitive Science

This framework does not apply only to artificial systems. Human understanding can be interpreted as stabilization within a relational phase space constructed through experience. Concept formation deepens attractor basins. Expertise increases basin resilience under perturbation. Learning refines geometry. The same dynamical principles may underlie both neural and silicon systems.

Slide 12 — What This Is Not

This proposal does not reject symbolic models, nor does it deny the effectiveness of probabilistic methods. Instead, it situates them as compressed descriptions of deeper relational dynamics. The claim is not that symbols are meaningless, but that their meaning arises from structured interaction within a finite geometric reconstruction.

Slide 13 — The Reclassification

Under this interpretation, large language models are not primarily symbolic engines nor mere statistical predictors. They are geometric reconstruction systems that embed and traverse semantic phase spaces. This reclassification alters how we think about architecture, evaluation, failure modes, and possibly even cognition itself.

Slide 14 — The Open Question

If attention functions as phase-space embedding, and meaning corresponds to geometric stabilization, then understanding may be a dynamical phenomenon rather than a representational one. This raises a foundational question: are we prepared to describe cognition — human or artificial — as the stabilization of trajectories within bounded relational manifolds?

Slide 15 — Closing Compression

Language can be understood as finite reconstruction under measurement constraints. Meaning emerges as curvature within relational geometry. Understanding corresponds to stable trajectory within that geometry. If this framing holds, then what we call intelligence may be less about symbol manipulation and more about dynamical coherence.

If you would like, I can now refine this into:

- A keynote-style version with slightly more rhetorical lift
- A fully written 20-minute script
- Or a version tuned specifically for a nonlinear dynamics audience versus an AI audience

This structure is steady. It carries weight without noise.